

Week 6

Gauss-Jordan elimination (Section 3.3)

$Ax = b \rightarrow Rx = c$ with R in reduced row echelon form (RREF); works for *every* system!

2 3 6 8

	1	0		0	0		
		1		0	0		
				1	0		
					1		

RREF(2, 3, 6, 8), $r = 4$

In general: RREF($j_1, j_2, \dots, j_r$):

$I (m \times m)$: in RREF(1, 2, $\dots$, m)

$0 (m \times m)$: in RREF() ($r = 0$)

$j_1 j_2 j_3 j_4$

	1	0		0	0		
		1		0	0		
$i = 3$			$j < j_3$	1	0		
					1		

$e_1 e_2 e_3 e_4$

$j_1, j_2, \dots, j_r$ the “downward step” columns

(Definition 3.13)

$r_{ij} = 0$ whenever

$i \leq r$ and $j < j_i$

or

$i > r$

Lemma 3.14: R in RREF($j_1, j_2, \dots, j_r$) has independent columns $j_1, j_2, \dots, j_r$ and rank r .

“Proof” by picture:

independent $\downarrow$: only 0's left of 1

	1	0		0	0		
		1		0	0		
				1	0		
					1		

$e_1 e_2 e_3 e_4$

dependent $\downarrow$

	1	0		0	v_1	0	
		1		0	v_2	0	
				1	v_3	0	
						1	

$e_1 e_2 e_3 v e_4$

$$v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ 0 \\ 0 \\ 0 \end{pmatrix} = v_1 e_1 + v_2 e_2 + v_3 e_3$$

Direct solution of $Rx = c$ with R in RREF($j_1, j_2, \dots, j_r$):

$$\begin{array}{cccc}
 j_1 & j_2 & j_3 & j_4 \\
 \begin{array}{|c|c|c|c|c|c|c|c|}
 \hline
 1 & 0 & & 0 & 0 & & & \\
 \hline
 & 1 & & 0 & 0 & & & \\
 \hline
 & & & 1 & 0 & & & \\
 \hline
 & & & & 1 & & & \\
 \hline
 & & & & & & & \\
 \hline
 & & & & & & & \\
 \hline
 \end{array}
 & = &
 \begin{array}{|c|}
 \hline
 c_1 \\
 \hline
 c_2 \\
 \hline
 0 \\
 \hline
 c_3 \\
 \hline
 c_4 \\
 \hline
 0 \\
 \hline
 0 \\
 \hline
 0 \\
 \hline
 0 \\
 \hline
 \end{array}
 & = &
 \begin{array}{|c|}
 \hline
 c_1 \\
 \hline
 c_2 \\
 \hline
 c_3 \\
 \hline
 c_4 \\
 \hline
 0 \\
 \hline
 0 \\
 \hline
 0 \\
 \hline
 \end{array}
 \\
 R & & c & & x
 \end{array}$$

← if $\neq 0$ here, no solution

$$x_j = \begin{cases} c_i, & \text{if } j = j_i \\ 0, & \text{otherwise.} \end{cases}$$

canonical solution

Elimination: if A is not in RREF

- $Ax = b \rightarrow Rx = c$ (same solutions, R in RREF) focus on $A \rightarrow R$ below
- For $Rx = c$, apply direct solution

Like Gauss, except... turn pivots into 1, also eliminate *above* the pivots: Column 1

$$\begin{array}{l}
 \begin{bmatrix} \mathbf{2} & 4 & 2 & 2 & -2 \\ 6 & 12 & 6 & 7 & 1 \\ 4 & 8 & 2 & 2 & 6 \end{bmatrix} \\
 \text{divide (row 1) by 2:} \\
 \begin{bmatrix} \mathbf{1} & 2 & 1 & 1 & -1 \\ 6 & 12 & 6 & 7 & 1 \\ 4 & 8 & 2 & 2 & 6 \end{bmatrix} \\
 \text{subtract } 6 \cdot (\text{row 1}) \text{ from (row 2):} \\
 \begin{bmatrix} \mathbf{1} & 2 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 7 \\ 4 & 8 & 2 & 2 & 6 \end{bmatrix} \\
 \text{subtract } 4 \cdot (\text{row 1}) \text{ from (row 3):} \\
 \begin{bmatrix} \mathbf{1} & 2 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & -2 & -2 & 10 \end{bmatrix}
 \end{array}$$

Pivot **0** and no row exchange possible.

Column 2 $\rightarrow$ Column 3

$$\begin{array}{l}
 \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & \mathbf{0} & 0 & 1 & 7 \\ 0 & \mathbf{0} & -2 & -2 & 10 \end{bmatrix} \\
 \downarrow \\
 \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & \mathbf{0} & 1 & 7 \\ 0 & 0 & -2 & -2 & 10 \end{bmatrix}
 \end{array}$$

Case of failure in Gauss elimination. Here: case that saves us some work!

Column 3

$$\begin{array}{l}
 \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & \mathbf{0} & 1 & 7 \\ 0 & 0 & -2 & -2 & 10 \end{bmatrix} \\
 \text{exchange (row 2) and (row 3):} \quad \downarrow \\
 \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & -\mathbf{2} & -2 & 10 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix} \\
 \text{divide (row 2) by } -2: \quad \downarrow \\
 \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & \mathbf{1} & 1 & -5 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix} \\
 \text{subtract } 1 \cdot (\text{row 2}) \text{ from (row 1):} \quad \downarrow \\
 \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & \mathbf{1} & 1 & -5 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix}
 \end{array}$$

Column 4

$$\begin{array}{l}
 \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & \mathbf{1} & 7 \end{bmatrix} \\
 \text{subtract } 1 \cdot (\text{row 3}) \text{ from (row 2):} \quad \downarrow \\
 R = \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -12 \\ 0 & 0 & 0 & \mathbf{1} & 7 \end{bmatrix}
 \end{array}$$

What about the right-hand side b?

Very useful: Version with m right-hand sides as input (columns of $m \times m$ matrix B) and the m transformed right hand-sides as output (columns of C).

Algorithm 6:

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1: function GAUSS-JORDAN ELIMINATION( $A, B$ ) ▷  $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{m \times m}$ 
2:    $R \leftarrow A, C \leftarrow B, r \leftarrow 0$  ▷  $r$ : number of downward steps so far
3:   for  $j = 1, 2, \dots, n$  do ▷ eliminate in column  $j$ 
4:     if  $r = m$  then ▷ no further downward steps possible, done!
5:       break ▷ ... loop execution and go to line 21
6:     end if
7:      $s \leftarrow r + 1$  ▷ row of (potential) next downward step
8:     if  $r_{sj} = 0$  then ▷ zero pivot
9:       if there is some  $k > s$  such that  $r_{kj} \neq 0$  then
10:        exchange (row  $s$ ) and (row  $k$ ) (in both  $R$  and  $C$ ) ▷ row operation

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11:         else                                ▷ no downward step in column  $j$ 
12:         continue                            ▷ ... in line 3 with the next column
13:     end if
14: end if                                     ▷ now  $r_{sj} \neq 0$ 
15: divide (row  $s$ ) by  $r_{sj}$  (in both  $R$  and  $C$ )    ▷ row operation
16: for  $i = 1, 2, \dots, s-1$  and  $i = s+1, s+2, \dots, m$  do    ▷ make  $r_{ij} = 0$ 
17:     subtract  $r_{ij} \cdot$  (row  $s$ ) from (row  $i$ ) (in both  $R$  and  $C$ )    ▷ row operation
18: end for                                    ▷ now, the  $j$ -th column of  $R$  equals  $e_s$ 
19:  $r \leftarrow r + 1, j_r \leftarrow j$           ▷ next downward step was made in column  $j$ 
20: end for
21: return  $(R, j_1, j_2, \dots, j_r, C)$         ▷  $R$  is in RREF( $j_1, j_2, \dots, j_r$ )
22: end function

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Runtime for $(A, B) \rightarrow (R, C)$:

- $O(m)$ row operations in each of the r downward steps ($r \leq m$)
- $O(m+n)$ basic operations per row operation (C : m columns, A : n columns)
- Time $O(mr(m+n))$ in total (also covers the other operations)

For $m = r = n$ (Gauss success scenario): $O(m^3)$.

Theorem 3.17: Let A be an $m \times n$ matrix, and let $(R, j_1, j_2, \dots, j_r, M)$ be the output of Algorithm 6 with input (A, I) . Then M is invertible, $R = MA$, and R is in RREF($j_1, j_2, \dots, j_r$).

Proof. $\underbrace{C}_M = \underbrace{M_\ell M_{\ell-1} \dots M_1}_{\text{row operation matrices}} \underbrace{B}_I \quad \Rightarrow \quad R = \underbrace{M_\ell M_{\ell-1} \dots M_1}_M A.$

Since all M_i are invertible (row operations are undoable), their product M is also invertible (Lemma 2.59). $\square$

R is the “RREF standard form” of A and also gives us the CR decomposition:

Theorem 3.18: Let A be an $m \times n$ matrix. There is a unique $m \times n$ matrix R (the one resulting from Gauss-Jordan elimination; Theorem 3.17), with the following two properties.

- (i) $R = MA$ for some invertible $m \times m$ matrix M .
- (ii) R is in RREF.

More precisely, R is in RREF($j_1, j_2, \dots, j_r$), where $j_1, j_2, \dots, j_r$ are the indices of the independent columns in A , and

$$R = \left[\begin{array}{c} \underbrace{R'}_{r \times n} \\ \underbrace{0}_{(m-r) \times n} \end{array} \right],$$

with R' the unique matrix such that $A = CR'$ in Theorem 2.46 (CR decomposition).

Verify this on

$$\underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}}_C \underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}}_{R'}$$

from Section 2.3.5 by doing Gauss-Jordan on A :

$$\begin{array}{l} A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \\ \text{elimination in column 1:} \\ \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 2 & -4 \end{bmatrix} \downarrow \\ \text{elimination in column 3:} \\ \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \downarrow \\ R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Computing the inverse of A : Run Gauss-Jordan on $(A, I) \rightarrow (R, M)$.

Theorem 3.19: A invertible $\Leftrightarrow R = I$ (and then $M = A^{-1}$).

Solving $Ax = b$ (possibly for many b): Run Gauss-Jordan on $(A, I) \rightarrow (R, M)$.

Theorem 3.20: Set $c = Mb$ and solve $Rx = c$ ($MAx = Mb$) by direct solution.

$O(m^2)$ time per b , after $O(mr(m+n))$ time for Gauss-Jordan.

Vector spaces (Section 4.1)

https://ti.inf.ethz.ch/ew/courses/LA25/slides/vectors_handout.pdf

Bases and dimension (Section 4.2)

Basis of a vector space V : linearly independent vectors that span V .

Need linear combinations / span in vector spaces.

Previously, we used *sequences* of vectors; now, we also work with *sets* (possibly infinite: polynomials).

Definition 4.15 Let V be a vector space, $G \subseteq V$ a (possibly infinite) subset of vectors. A *linear combination* of G is a sum of the form

$$\sum_{j=1}^n \lambda_j \mathbf{v}_j,$$

where $F = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a finite subset of G and $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$.

Lemma 4.16: Let V be a vector space. Every linear combination of V is again in V .

Proof is easy (linear combination is “natural behavior”, combining “add up” and “scale”).
But needs *finite* linear combinations:

Consider $\mathbb{R}[x]$ (polynomials), and the infinite “linear combination”

$$\sum_{j=0}^{\infty} x^j \left(= \frac{1}{1-x} \right).$$

of the *unit monomials* $1, x, x^2, \dots$. This is not a polynomial.

Definition 4.17: Let V be a vector space, $G \subseteq V$ (possibly infinite).

G is called *linearly dependent* if there is an element $\mathbf{v} \in G$ such that $\mathbf{v}$ is a linear combination of $G \setminus \{\mathbf{v}\}$. Otherwise, G is called *linearly independent*.

The *span* of G , $\text{Span}(G)$, is the set of all linear combinations of G .

Bases

Definition 4.18: Let V be a vector space. A subset $B \subseteq V$ is called a *basis* of V if B is linearly independent and $\text{Span}(B) = V$.

Examples: (For linear independence, use *private nonzero* argument!)

vector space V	basis B
$\mathbb{R}^m$	$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$
$C(A)$ (subspace of $\mathbb{R}^m$)	independent columns of A
2×2 symmetric matrices (subspace of $\mathbb{R}^{2 \times 2}$)	$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
$\mathbb{R}[x]$ (polynomials)	$\{x^i : i = 0, 1, \dots\}$ (infinite set)
$\{0\}$ (smallest vector space)	$\emptyset$ (empty set)

There can be many bases:

Observation 4.20: Every set $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ of m linearly independent vectors is a basis of $\mathbb{R}^m$.

Proof. B is linearly independent, and $\text{Span}(B) = \mathbb{R}^m$ is true by Lemma 1.28. □

Does every vector space have a basis? Yes, but we only treat the *finitely generated* case.

Definition 4.21: A vector space V is called *finitely generated* if there exists a finite subset $G \subseteq V$ with $\text{Span}(G) = V$.

$\mathbb{R}^m$ is finitely generated (by $G = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$) but $\mathbb{R}[x]$ is not.

Theorem 4.22: Let V be a finitely generated vector space, and $G \subseteq V$ a finite subset with $\text{Span}(G) = V$. Then V has a basis $B \subseteq G$.

(Algorithmic) Proof. Start with G .

1. If G is linearly independent, G is a basis.
2. Otherwise, some $v \in G$ is a linear combination of $G \setminus \{v\}$ (Definition 4.17) and $\text{Span}(G \setminus \{v\}) = \text{Span}(G) = V$ (Corollary 1.27).
3. Replace G with $G \setminus \{v\}$ and goto 1.

G is finite and gets smaller in every round, so this must stop with a basis. $\square$

The Steinitz exchange lemma

Lemma 4.23: Let V be a finitely generated vector space, $F \subseteq V$ a finite set of linearly independent vectors, and $G \subseteq V$ a finite set of vectors with $\text{Span}(G) = V$. Then the following two statements hold.

- (i) $|F| \leq |G|$.
 - (ii) There exists a subset $E \subseteq G$ of size $|G| - |F|$ such that $\text{Span}(F \cup E) = V$.
- (ii) means: can enlarge F by some elements from G such that the result has at most the size of G and also spans V .

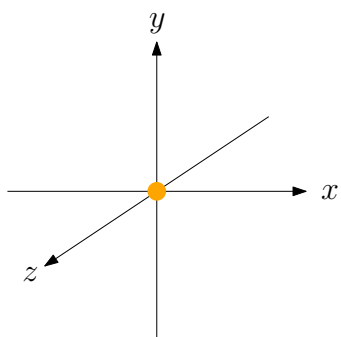
Lemma 1.28 is the “poor man’s” version of this. Proof for both is (almost) the same (Exercises).

Theorem 4.24 Let V be a finitely generated vector space and let $B, B' \subseteq V$ be two bases of V . Then $|B| = |B'|$.

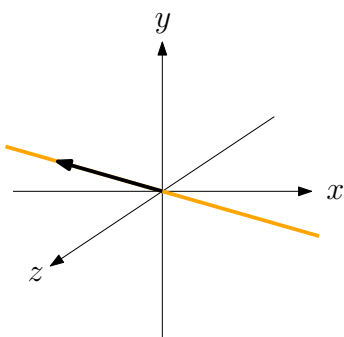
Proof. B and B' are linearly independent, and $\text{Span}(B) = \text{Span}(B') = V$ (Definition 4.18). Steinitz exchange Lemma 4.23 (i) with $F = B, G = B'$ gives $|B| \leq |B'|$; with $F = B', G = B$, we get $|B'| \leq |B|$. $\square$

Dimension

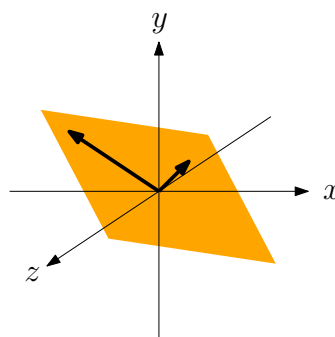
Definition 4.25: Let V be a finitely generated vector space. Then the *dimension* of V , $\dim(V)$, is the size of an arbitrary basis B of V .



dimension 0



dimension 1



dimension 2

Now we can finally say: $\mathbb{R}^m$ has dimension m .

:-)